13. Scenario: You are working on a text analysis project and need to determine the frequency distribution of words in a given text document. You have a text document named "sample_text.txt" containing a paragraph of text. Your task is to develop a Python program that reads the text document, processes the text, and generates a frequency distribution of the words.

Question: How would you develop a Python program to calculate the frequency distribution of words in a text document?

```
from collections import Counter

import re

# Read the text file

with open("sample_text.txt", "r") as file:

  text = file.read()

# Convert to lowercase and extract words

words = re.findall(r'\b\w+\b', text.lower())

# Calculate frequency distribution

frequency = Counter(words)

print("Word Frequency Distribution:")

for word, count in frequency.items():

  print(word, ":", count)
```

```
============== RESTART: C:/Users/Jannathul firdaus/exp 14 fods.py ==============
Word Frequency Distribution:
 python : 3
 is : 3
 easy : 1
 powerful : 1
 useful : 1
```

14. Scenario: You are a data analyst working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas data frame.

Question: Develop a code in python to find the frequency distribution of the ages of the customers who have made a purchase in the past month.

```
import pandas as pd

# Create customer purchase data

data = {
```

```
    'Age': [25, 30, 25, 40, 30, 25, 35],
    'Purchase_Date': [
        '2026-10-02', '2026-10-04', '2026-10-07',
        '2026-10-10', '2026-10-12', '2026-10-15',
        '2026-10-18'
    ]
}
df = pd.DataFrame(data)
df['Purchase_Date'] = pd.to_datetime(df['Purchase_Date'])

# Filter purchases for October 2026
monthly_data = df[
    (df['Purchase_Date'] >= '2026-10-01') &
    (df['Purchase_Date'] < '2026-11-01')
]
# Calculate age frequency distribution
frequency = monthly_data['Age'].value_counts().sort_index()
print("Customer Age Frequency Distribution:")
print(frequency)
```

```
============== RESTART: C:/Users/Jannathul firdaus/exp 14 fods.py ==============
Customer Age Frequency Distribution:
Age
25    3
30    2
35    1
40    1
Name: count, dtype: int64
```

15. Scenario: You are a data analyst working for a social media platform. As part of your analysis, you have a dataset containing user interaction data, including the number of likes received by each post. Your task is to develop a Python program that calculates the frequency distribution of likes among the posts.

Question: Develop a Python program to calculate the frequency distribution of likes among the posts?

```
import pandas as pd

# Create post interaction data

data = {

   'Post_ID': [1, 2, 3, 4, 5, 6, 7, 8],

   'Likes': [100, 200, 100, 300, 200, 100, 400, 200]

}

df = pd.DataFrame(data)

# Calculate frequency distribution of likes

frequency = df['Likes'].value_counts().sort_index()

print("Frequency Distribution of Likes:")

print(frequency)
```

```
========== RESTART: C:/Users/Jannathul firdaus/exp 14 fods.py ==========
Frequency Distribution of Likes:
Likes
100    3
200    3
300    1
400    1
Name: count, dtype: int64
```